Caleb R. Pucylowski

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Education

Duke University — Durham, NC
B.S.E. Electrical & Computer Engineering and Computer Science (Double Major) — *Expected May 2028*
- Minor: Mathematics | Concentration: Artificial Intelligence & Machine Learning | Dean's List with Distinction (Fall 2024)
- Coursework: Data Structures & Algorithms, Algorithms, Operating Systems, Software Engineering, Computer Architecture, Networks, Signals & Systems | In Progress: Applied Machine Learning, Robotics, Game Design

Eastern Florida State College — Melbourne, FL
Associate of Arts | GPA: 4.0/4.0 | 68 college credits earned through dual enrollment in high school — *May 2024*

Experience

Software Engineering Intern — June 2026 – Aug. 2026
NASA Langley Research Center, Simulation Development & Analysis Branch — *Hampton, VA*
- Built an in-house **C++** "Highway in the Sky" 3D flight-guidance display for a NASA urban air mobility (UAM) aircraft simulation with **OpenSceneGraph**/**OpenGL**, replacing third-party vendor graphics that were hard to modify
- Turned waypoint data (latitude, longitude, altitude, heading) into real-time guidance on the primary-flight and navigation displays; designed the data structures, display flags, and **XML** configuration
- Debugged 3D coordinate transforms, pitch/roll/yaw, unit conversion, and terrain overlays on **Linux** and **Windows**; flight tests on NASA's full-scale flight-deck simulator matched the vendor display on every route
- Finished core development **2 weeks ahead of schedule**; passed critical design review, live demo, and code review

Software Development Engineer Intern — May 2025 – Aug. 2025
Amazon, Private Brands — *Seattle, WA*
- Built an end-to-end escalation service for an AI shopping assistant in **Java** and **TypeScript**, using **WebSockets** to trigger escalations in real time and route them to subject-matter experts (SMEs)
- Cut SME case-review time by **83%** (30 min to 5 min) by attaching AI-generated summaries and full conversation history to each ticket, freeing capacity for **4–6 more escalations per reviewer per day**
- Integrated **AWS Lambda**, **API Gateway** (**GraphQL**), and **DynamoDB** with least-privilege **IAM** roles; wrote unit tests so **100%** of escalations reached reviewers with complete context
- Authored the technical design doc, demoed to stakeholders, and shipped via Agile sprints and code reviews

Undergraduate Research Assistant — May 2025 – May 2026
Duke Quantum Center — *Durham, NC*
- Developed and simulated quantum circuits in **Python** (**Qiskit**, **Cirq**); analyzed error on noisy quantum hardware

Research Assistant — June 2024 – Aug. 2024
University of Maryland, Baltimore County (Deffner Lab) — *Baltimore, MD*
- Ran noisy quantum-system simulations to quantify error and decoherence in open quantum systems

Projects

Bazaar Multiplayer Game Engine | *Java, Gradle, JUnit 5, Gson* — Spring 2026
- Co-developed a **Java** multiplayer game engine (rules engine, referee, pluggable AI players) with **226 JUnit 5 tests**

LLM Fine-Tuning & Retrieval-Augmented Generation | *Python, PyTorch, CUDA, Hugging Face, LoRA* — Fall 2026
- Fine-tuned a 1.3B-parameter LLM (Phi-1.5) on GPUs with **CUDA** and **LoRA**, reaching **98%** instruction-following on GSM8K
- Built a **RAG** pipeline over 10,570 passages (**9.6 ms** vector search per query), raising LLM QA accuracy from 21% to **58%**

Autonomous Patrol Robot | *ROS 2, Python, Kinova Robotic Arm* — Fall 2026 (In Progress)
- Building a **ROS 2** robot in **Python** with a modified Kinova arm that patrols and warns off people it detects nearby

Vengeful Glory: 2D Dungeon Crawler | *Unity, C#, Team of 3* — Fall 2026 (In Progress)
- Co-developing a game where a defeated boss levels up to take revenge: XP/item progression, enemy AI, gated zones

Operating Systems & Computer Architecture Projects | *C, RISC-V, pthreads, Logisim* — 2025
- Implemented kernel features in **C** for the xv6 teaching OS on **RISC-V** (traps, virtual memory, threads, file system); built a heap allocator with block splitting and coalescing, and a priority reader-writer lock with **pthreads**
- Designed a **16-bit CPU** from logic gates: 8-register file, ALU, program counter, decoder, memory, and I/O

Technical Skills

Languages: C, C++, Python, Java, C#, TypeScript, JavaScript, Lua, SQL, Bash, MATLAB, R, HTML/CSS, XML
Cloud & Backend: AWS (Lambda, API Gateway, DynamoDB, S3, EC2, IAM, Bedrock), REST, GraphQL, WebSockets
Systems & Graphics: Linux, ROS 2, Unity, Arduino, RISC-V/xv6, OpenSceneGraph, OpenGL, Real-Time 3D Simulation
AI/ML & Quantum: PyTorch, CUDA, Hugging Face, NumPy, SciPy, Pandas, Jupyter, LLMs, Qiskit, Cirq, D-Wave Ocean (QUBO)
Tools & Practices: Git/GitHub, Docker, Gradle, JUnit, ClearCase, LabVIEW, KiCad, VS Code, Code Review, Agile/Scrum

Honors & Leadership

Honors: QuestBridge Match Scholar; Generation Google Scholar; Amazon Future Engineer Scholar; Hispanic Scholarship Fund Scholar; MIT MITES Semester Scholar; National Merit Commended Scholar; National Cyber League Top 3%
Leadership & Activities: Duke Engineering Student Government, VP of Marketing & Communications; MATLAB Club, Co-Founder & Treasurer; FIRST Robotics, Programmer & Safety Captain (4 yrs); Duke Club Rowing